

ELENA JOHANNESSEN

Data Analyst | ETL, BI & LLM-Driven Analytics

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WORK EXPERIENCE — DATA ANALYTICS

Data Analyst

October 2025 – Present

Zoca AI — San Jose, CA (Remote)

- Built an end-to-end conversation analytics pipeline using Python and SQL to reconstruct and analyze customer chat interactions, enabling session-level insights across thousands of messages.
- Applied LLM-based analysis to identify customer intent, drop-off points, and chatbot failure patterns, generating actionable insights to improve booking conversion rates and reduce manual handovers.

Data Analyst

January 2022 – September 2025

Hamilton Practice — San Francisco, CA (Remote)

- Designed and managed 44 ETL pipelines to extract data from Athena Health (EMR), ADT feeds, and OneDrive files, normalizing and storing them in RDS databases with built-in check constraints to enforce data quality.
- Created and maintained ChatGPT prompts to accelerate ad-hoc reports and standard BI dashboards, reducing build time from 20 hours to under 1 hour.
- Monitored and analyzed costs across AWS serverless stacks, Athena Health, and other cloud platforms, detecting anomalies that prevented a 1000% cost increase.
- Optimized database query performance by tracking execution of top queries and implementing indexes, reducing response times and enhancing system efficiency.

DATA ANALYTICS & AI PROJECTS

LLM-Based Emotional State Analysis for Botanical Wellness Interventions — CrowdDoing (Nonprofit)

- Built an LLM-powered analysis pipeline using GPT-2 OSS models to evaluate pre- and post-intervention emotional states from user feedback on botanical treatments, mapping qualitative text to structured scoring metrics.
- Designed a scoring framework comparing emoji-based self-reported emotions with post-experience narratives, identifying sentiment shifts and quantifying alignment between expected and actual outcomes.

Streamlit AI Assistant — Share Price & SEC Filings Analysis

- Developed a Streamlit application enabling users to query stock data via natural language, using OpenAI's function-calling APIs to translate speech into API calls for share prices and recent SEC filings.
- Integrated financial APIs to retrieve stock data and SEC 8-K filings, supporting broad analysis of potential investments based on user input.
- Employed GPT-3.5-Turbo to convert user text into JSON for API calls, and GPT-4-Turbo to transform JSON API responses back into conversational text.

Streamlit AI Assistant — CFPB Database for Credit Products

- Developed a Streamlit application to enhance user interaction with an AI assistant, reducing task completion time by 15% compared to a Jupyter notebook workflow.
- Implemented the assistant using GPT-4 Turbo with a vector store managing seven data files, demonstrating a scalable approach to data processing.
- Constructed polling mechanisms that query the AI assistant every five seconds for responses and allow users to initiate new threads, keeping the interface responsive.

TECHNICAL SKILLS

Languages & Analysis: Python, SQL, SciPy, Scikit-learn, NumPy, Pandas, Jupyter Notebook, Advanced Excel

Data Engineering: ETL, Docker, Kafka, Postgres, Conda, Git, Microsoft Graph, GCP, Cloud

BI & Visualization: Tableau, Power BI, Metabase, Data Visualization

AI / ML & Tools: OpenAI APIs, Streamlit, Keras, TensorFlow, BeautifulSoup, Web Scraping

EDUCATION

MS, Data Science — Pace University, Seidenberg School of Computer Science

Sep 2022 – May 2024

GPA: 3.9

BBA, Finance — Pace University, Lubin School of Business

Sep 2017 – Dec 2021

GPA: 3.4, Dean's List